CHRIS DAVID NAKOJAH

Electrical Engineer

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EDUCATION

Michigan Technological University — M.S. Electrical & Computer Engineering *Expected Apr 2027*
Houghton, Michigan, USA

Université de Bourgogne Europe — M.S. Advanced Electronic Systems Engineering *Sept 2023 – Oct 2024*
Dijon, France

Kwame Nkrumah University of Science and Technology — B.S. Telecommunication Engineering *Aug 2018 – Sept 2022*
Kumasi, Ghana

TECHNICAL SKILLS

Control Systems Design: Dynamic System Modeling, Controller Design (PID, LQR, State-Feedback), Kalman Filtering
Automation & Industrial Controls: Ladder Logic, SCADA/HMI, Siemens & Allen-Bradley PLCs, TIA Portal, Studio 5000
Embedded Systems: Embedded C, RTOS, CAN, LIN, Arduino, ARM Cortex-M, SoCs, Zynq, Artix, STM32 Microcontrollers
Programming & Software: C, C++, Python, MATLAB, Simulink, VHDL, Verilog, LabVIEW, ModelSim, CANoe, CODESYS
Hardware & Digital Design: FPGA Design, Xilinx, Vivado, RTL Design, PCB Design, CAD Tools
Languages: English (Fluent), French (Fluent)

EXPERIENCE

Systems & Control Engineer, ISAE-SUPAERO *Jan 2025 – Jun 2025*
Complex Systems Engineering Department — Paris, France

- Implemented real-time control algorithms on STM32 microcontrollers using C and Simulink-generated code
- Integrated sensors, actuators, and motor drivers into closed-loop control systems
- Performed dynamic control simulation, modeling, and validation using MATLAB/Simulink and HIL testing

Embedded Systems Engineering Intern, ISAE-SUPAERO *Jun 2024 – Oct 2024*
Toulouse, France

- Designed an Ethernet gateway supporting 1 Gbps deterministic communication for the FactoRing project
- Implemented IEEE Time-Sensitive Networking architecture enabling low-latency deterministic Ethernet communication
- Integrated and validated the gateway on a Xilinx SoC-based ALINX AX9 platform through hardware verification testing

Graduate Research & Hardware Design, Université de Bourgogne Europe *Jan 2024 – May 2024*
Dijon, France

- Designed a cryptographic architecture to strengthen embedded hardware security against implementation-level attacks
- Developed an advanced encryption standard (AES) dynamic S-Box architecture that improved cryptographic robustness
- Implemented and verified the encryption architecture on a Xilinx FPGA platform using RTL design and hardware validation

PROJECTS

Autonomous Vehicle Parking System
Tools/Hardware: 3x Arduino Mega 2560, CAN Bus Transceivers, Ultrasonic Sensor, Servo Motor

- Designed a three-node CAN network integrating ultrasonic sensing, servo scanning, and motor actuation for autonomous parking
- Validated real-time obstacle-detection and motion-decision logic using read_dist(), servo control, and CAN motor commands

Automated Temperature-Controlled Conveyor Cell Using PLC
Tools/Hardware: Allen-Bradley PLC, Ladder Logic, Temperature & Proximity Sensors, Conveyor Motor

- Designed a PLC-controlled conveyor cell in ladder logic with temperature, part, and outfeed sensing for heat-cool automation
- Implemented the system in closed-loop temperature control with fault-alarm logic

Closed-Loop Motor Position Control
Tools/Hardware: MATLAB/Simulink, BeagleBone Blue (ARM Cortex-A8), Quadrature Encoder, PWM Motor Control

- Deployed proportional DC motor position control on a BeagleBone Blue via Simulink, using encoder feedback and PWM actuation
- Identified motor dynamics from measured voltage and position data, validating the model and tracking under actuator saturation